Funica Flutter App: Step by Step Roadmap

Furniture e-commerce app (Flutter + MVVM + Riverpod + Firebase). Ye roadmap setup se leke release tak ka poora plan hai.

Tech Stack

Cheez	Use
Framework	Flutter (Dart)
Architecture	MVVM (View, ViewModel, Repository)
State management	Riverpod
Navigation	go_router
Backend	Firebase (Auth, Firestore, Storage, Messaging, Functions)
Payments	Razorpay ya Stripe
Local secure data	flutter_secure_storage, local_auth

Phase 0: Environment Setup (Day 1)

1. Flutter SDK, Android Studio (emulator) aur VS Code (Flutter + Dart extensions) install karo.
2. `flutter doctor` chalao, jo bhi issue dikhe fix karo.
3. `create_funica_mvvm.ps1` chalao, project ban jayega.
4. Git repo banao: `git init`, GitHub pe push karo.
5. Firebase tools install karo:

```
npm install -g firebase-tools
dart pub global activate flutterfire_cli
firebase login
```

Phase 1: UI Foundation (Day 2-4)

1. `app_colors.dart` aur `app_theme.dart` final karo (dark theme, font Urbanist).
2. go_router setup karo (`app_router.dart`): saare routes define karo, aur bottom nav ke liye ShellRoute (Home, Wishlist, Cart, Orders, Profile).
3. Shared widgets banao: PrimaryButton, AppTextField, ProductCard, CategoryChip, SuccessDialog.
4. Splash aur Onboarding screens banao.
5. Is phase mein sab kuch dummy data se chalao, backend abhi nahi.

Phase 2: Firebase Setup (Day 4-5)

1. console.firebase.google.com pe naya project banao.
2. Project folder mein `flutterfire configure` chalao. Ye `firebase_options.dart` bana deta hai.
3. Packages add karo:

```
flutter pub add firebase_core firebase_auth cloud_firestore firebase_storage
firebase_messaging google_sign_in
```

4. `main.dart` mein `await Firebase.initializeApp(options: DefaultFirebaseOptions.currentPlatform);` lagao.
5. Console mein enable karo:
   - Authentication: Email/Password, Google, Phone (OTP ke liye)
   - Firestore Database (shuru mein test mode, baad mein rules lagana)
   - Storage (profile photo aur product images)
   - Cloud Messaging
6. Android ke liye SHA-1 key console mein add karo (Google Sign-In ke liye zaroori): `cd android` phir `./gradlew signingReport`.
7. Cloud Functions ke liye project ko Blaze plan pe upgrade karna padega (free quota milta hai, par card chahiye).

Phase 3: Firestore Data Design (Day 5-6)

Coding se pehle collections decide kar lo:

Collection	Fields (main)
`users/{uid}`	name, nickname, email, phone, dob, gender, country, photoUrl, fcmToken, createdAt
`users/{uid}/addresses`	label, fullAddress, lat, lng, isDefault
`users/{uid}/wishlist`	productId, addedAt
`users/{uid}/cart`	productId, qty, color
`users/{uid}/cards`	brand, last4, expiry (kabhi full card number store mat karo)
`products/{id}`	name, category, price, description, images[], colors[], rating, reviewCount, soldCount, stock
`products/{id}/reviews`	userId, userName, rating, comment, likes, createdAt
`categories/{id}`	name, iconUrl
`offers/{id}`	title, discountPercent, productId, bannerUrl, expiresAt
`promos/{id}`	code, discountType, value, minAmount, expiresAt
`orders/{id}`	userId, items[], addressId, shippingType, promoCode, total, status, paymentMethod, createdAt
`orders/{id}/tracking`	status, message, timestamp
`wallets/{uid}`	balance
`transactions/{id}`	userId, type (topup/purchase), amount, orderId, createdAt
`notifications/{id}`	userId, title, body, type, isRead, createdAt
`chats/{uid}/messages`	text, senderType, createdAt
`faqs/{id}`	question, answer, category

Products ka seed data Firestore console se ya ek chhoti script se daal do taaki UI test ho sake.

Phase 4: Auth Module (Week 1-2)

1. `AuthRepository` mein Firebase Auth ke methods likho: signUp, signIn, signOut, signInWithGoogle, sendOtp, verifyOtp, resetPassword.
2. `AuthViewModel` mein loading, error aur user state manage karo.
3. Screens connect karo: Let's you in, Signup/Login, Fill Profile, Create PIN, Fingerprint.
4. Fill Profile mein photo Firebase Storage pe upload karo aur data Firestore `users/{uid}` mein save karo.
5. Fingerprint ke liye `local_auth` package.
6. PIN ke liye `flutter_secure_storage` (hash karke store karo, plain nahi).
7. Forgot Password flow: Firebase `sendPasswordResetEmail` ya phone OTP, phir naya password.
8. App start pe `authStateChanges()` se decide karo ki user ko login screen dikhana hai ya Home.

Phase 5: Home, Categories aur Products (Week 2-3)

1. `ProductRepository` : getCategories, getProducts(category), getPopularProducts, getOffers, getProductById.
2. Home screen: greeting, search bar, Special Offers banner (PageView), 8 category icons, Most Popular grid with filter chips.
3. Category listing screens (Sofa, Chair, Table, Kitchen, Lamp, Cupboard, Vase) ek hi reusable screen banao jo category parameter le.
4. Product Detail: image carousel (`smooth_page_indicator`), color swatches, quantity stepper, price, Add to Cart.
5. Reviews screen: rating filter chips, list, likes.
6. Images ke liye `cached_network_image` .
7. Pagination ke liye Firestore `limit()` aur `startAfter()` use karo.

Phase 6: Search, Filter aur Sort (Week 3)

1. Recent searches local storage mein (`shared_preferences`).
2. Firestore mein full-text search nahi hota. Options: (a) chhota catalog ho to saare products fetch karke client pe filter karo, (b) Algolia ya Typesense lagao.

3. Sort and Filter bottom sheet: category, price range slider, rating, sort order, Reset/Apply.
4. Empty state (no results) screen.

Phase 7: Wishlist aur Cart (Week 3-4)

1. Wishlist: heart toggle, Firestore `users/{uid}/wishlist`, real-time `snapshots()` stream.
2. Cart: add, remove, quantity update, total calculation.
3. Remove from Cart confirmation sheet.
4. Cart badge bottom nav pe.
5. Logic ViewModel mein rakho, View mein nahi.

Phase 8: Checkout Flow (Week 4-5)

1. Address management: list, add (map ke saath `google_maps_flutter` ya `geolocator`), edit, default select.
2. Choose Shipping: Economy, Regular, Cargo, Express (price aur days config se).
3. Promo code: Firestore `promos` se validate karo, discount calculate karo.
4. Payment methods: Wallet, Razorpay/Stripe, Google Pay, Apple Pay, saved card.
5. Payment ka secure part Cloud Functions mein karo (order create, payment verify). Client pe amount ka bharosa mat karo, server pe price dobara calculate karo.
6. Enter PIN screen se payment confirm karo.
7. Order Successful sheet, phir View Order ya E-Receipt.

Phase 9: Orders aur Tracking (Week 5)

1. My Orders: Active aur Completed tabs.
2. Track Order: status timeline (Packing, Shipped, In Transit, Delivered) `orders/{id}/tracking` se.
3. Status update admin side se ya Cloud Function se hoga, aur har update pe FCM notification bhejo.
4. Leave a Review: rating stars + comment, `products/{id}/reviews` mein save, aur product ka average rating update karo (Cloud Function ya transaction se).
5. E-Receipt screen: barcode (`barcode_widget`), amount, tax, payment details.

Phase 10: E-Wallet (Week 5-6)

1. `wallets/{uid}` balance dikhao, transaction history `transactions` se.
2. Top Up: amount select, payment method, PIN, success.
3. Balance update sirf Cloud Function se karo (client ko direct write permission mat do), warna koi bhi apna balance badha sakta hai.
4. Transaction History list aur E-Receipt.

Phase 11: Profile aur Settings (Week 6)

1. Profile screen aur Edit Profile.
2. Address list aur Add New Address.
3. Notification settings toggles (Firestore ya local mein save).
4. Payment settings aur Add New Card (sirf last4 aur brand store karo, card number payment gateway ke SDK se handle hoga).
5. Security: Face ID/Biometric toggle, Change PIN, Change Password.
6. Language: `flutter_localizations` aur `intl` se. Dark mode toggle `ThemeMode` se.
7. Privacy Policy (static text), Invite Friends (`share_plus` ya `contacts_service`), Logout.

Phase 12: Help Center aur Chat (Week 6-7)

1. FAQ accordion `faqs` collection se, search ke saath.
2. Contact Us options: WhatsApp, Website, Facebook, Instagram (`url_launcher`).
3. Customer Service chat: Firestore real-time `chats/{uid}/messages` . Simple start ke liye ek-tarfa support inbox kaafi hai.

Phase 13: Push Notifications (Week 7)

1. `firebase_messaging` setup, permission maango, FCM token `users/{uid}` mein save karo.
2. Foreground notifications ke liye `flutter_local_notifications` .
3. Cloud Functions se trigger bhejo: order status change, offers, payment success.
4. In-app Notification screen `notifications` collection se (Today, Yesterday, Older grouping).

Phase 14: Security Rules aur Backend Hardening (Week 7-8)

1. Firestore rules likho: user sirf apna data padh/likh sake, products sabke liye read-only, orders create sirf logged-in user, wallet aur transactions client se write band.
2. Storage rules: user sirf apne folder mein upload kare, file size aur type limit.
3. Firebase App Check enable karo.
4. Test mode rules production se pehle hata do.
5. Sensitive logic (payment, wallet, stock kam karna) Cloud Functions mein rakho.

Phase 15: Testing aur Polish (Week 8)

1. Unit tests: ViewModels aur Repositories (mock repository ke saath).
2. Widget tests: login, cart, checkout.
3. Real device pe test: fingerprint, notifications, payments.
4. Loading states (shimmer), error states, empty states har screen pe.
5. Offline handling aur no-internet message.
6. Performance: image sizes, list pagination, `const` widgets.
7. Crash reporting ke liye Firebase Crashlytics aur analytics ke liye Firebase Analytics.

Phase 16: Release (Week 9)

1. App icon aur splash: `flutter_launcher_icons`, `flutter_native_splash`.
2. App name, package id (`com.funica.app`), version set karo.
3. Android: keystore banao, `key.properties`, `flutter build appbundle --release`.
4. Google Play Console account (one-time fee), store listing, screenshots, privacy policy URL.
5. iOS ke liye Mac aur Apple Developer account chahiye, Android pehle release karna aasan hai.
6. Internal testing track pe pehle bhejo, phir production.

Suggested Timeline

Week	Kaam
1	Setup, UI foundation, Firebase, Firestore design
2	Auth, Home, Products
3	Search/Filter, Wishlist, Cart
4-5	Checkout, Orders, Tracking
5-6	Wallet, Profile, Settings
7	Help, Chat, Notifications
8	Security rules, Testing, Polish
9	Release

Tips

- Ek time pe ek feature poora karo (UI se backend tak), phir agla.
- Pehle UI dummy data se banao, phir Repository mein Firebase connect karo. MVVM mein ye aasan hai kyunki sirf Repository badalta hai.
- Har feature ke baad git commit karo.
- Pehle Auth, Home aur Cart ko solid karo, baaki features baad mein.
- Payment aur wallet ke liye real money test karne se pehle gateway ka test mode use karo.